



NETWORK & CRYPTOGRAPHIC SECURITY

Network Security Policy

Documents the air-gapped architecture and the subprocessor-provided network controls. Masar Zero does not operate its own firewalls, IDS/IPS, or VPN.

This policy covers network architecture, segmentation, boundary protection, remote access, wireless, ACLs, encryption, and capacity management. The air-gap from the customer network is the primary network security control. Cloud-side network security is delegated to Vercel, Supabase, and WorkOS platform features.

PREPARED BY	Masar Zero W.L.L.
DOCUMENT ID	MZ-POL-006
VERSION	v1.0
EFFECTIVE	12 July 2026
CLASSIFICATION	Internal Use — Authorized Recipients Only

Document Control

Document ID	MZ-POL-006	Version	v1.0
Effective Date	12 July 2026	Review Cycle	Annual (next review: July 2027)
Document Owner	Hari Krishna	Classification	Internal Use
Publisher	Masar Zero W.L.L.	Contact	hari@masarzero.com

Revision History

Version	Date	Author	Summary of Change
v1.0	12 July 2026	Masar Zero	Initial release of Network Security Policy.

Approval

This document is prepared by Hari Krishna and approved by Dr. A. Latif Alkhaja, Chairman of Masar Zero W.L.L. It is effective as of the date shown above and is reviewed annually or upon significant change per MZ-POL-001 §7. Distribution is restricted to authorized recipients; redistribution requires written approval from Masar Zero leadership.

Role	Name	Signature	Date
Prepared by	Hari Krishna		
Approved by	Dr. A. Latif Alkhaja (Chairman)		

1. Purpose & Scope

This policy defines the network security controls Masar Zero W.L.L. applies across its deployment: (a) the plant-floor equipment at customer service stations (WeCon box + router); (b) the WeCon cloud platform (vendor-operated); (c) the Masar Zero cloud environment hosted on Vercel, Supabase, and WorkOS.

The fundamental architectural assumption is that the Masar Zero deployment is air-gapped from the customer corporate network. The station-side router connects to an external Wi-Fi or cellular network (separate from the customer's network) to send telemetry to the WeCon cloud. This air-gap is documented in BAPCO-QR-001 Section 2 and is the justification for several N/A responses in the questionnaire.

2. Network Architecture

The Masar Zero network architecture is simple and relies heavily on subprocessor platform controls:

Tier	Components	Network Connection
Plant floor	VRU + WeCon V-Net box + router	Router connects to external Wi-Fi/cellular (NOT customer network)
WeCon cloud	Telemetry platform (vendor-operated)	WeCon-operated infrastructure
Masar Zero cloud	Vercel dashboard, Supabase DB, WorkOS auth	Subprocessor-managed (Vercel/Supabase/WorkOS platforms)
Founder laptops	macOS/Windows laptops	Internet via home/office Wi-Fi or cellular

Table 1 — Network tiers and connections.

3. Network Segmentation (Air-Gap)

Segmentation is enforced through three independent controls: (a) the station-side router connects to an external Wi-Fi or cellular network that is physically and logically separate from the customer station's network; (b) the WeCon box transmits telemetry only to the WeCon cloud platform (Masar Zero does not open any inbound listeners on station-side equipment); (c) the customer station's point-of-sale and inventory networks are on separate physical cabling and never connect to the Masar Zero router.

Air-gap enforcement. The combination of (a) external Wi-Fi/cellular network, (b) outbound-only telemetry to WeCon cloud, and (c) physical cable separation provides three independent controls against any traffic flow between the Masar Zero deployment and the customer corporate network. This is the architectural foundation on which several N/A questionnaire responses rely (see BAPCO-QR-001 §2.2).

4. Boundary Protection

The station-side router is the only network boundary on the plant floor. It uses manufacturer-default firewall settings. Masar Zero does not operate its own boundary firewalls in production. Cloud-side boundary protection is delegated to subprocessors: Vercel platform WAF and DDoS protection at the cloud edge; Supabase row-level security at the data layer; WorkOS at the auth layer.

5. Remote Access

Masar Zero defines two approved remote access methods:

- **Operator dashboard access:** WorkOS-authenticated HTTPS to the Masar Zero dashboard at app.masarzero.com (hosted on Vercel). Requires email + password + TOTP MFA. Session lifetime per WorkOS defaults.
- **Subprocessor admin console access:** Founders log in to Vercel, Supabase, WorkOS, WeCon cloud, Google Workspace, GitHub, and 1Password admin consoles from their laptops. Each requires 2FA.

All other remote access methods are prohibited. No VPN is in use (HTTPS is the remote access channel). No remote desktop protocol is permitted. WeCon box configuration is performed only from the local HMI at the station or via the WeCon cloud console.

6. Wireless Networking

The station-side router connects to an external Wi-Fi network (managed by the station operator or carrier) or uses a cellular SIM. Masar Zero does not deploy or manage Wi-Fi on the plant floor. The WeCon box and router do not have built-in Wi-Fi radios that Masar Zero configures. Founder laptops use standard home/office Wi-Fi or cellular.

Masar Zero does not perform RF sweeps for rogue access points. This is a scope limitation acknowledged in the master questionnaire response.

7. Cryptographic Protections

Masar Zero applies cryptographic protections through subprocessor platform features. Masar Zero does not manage TLS configuration or encryption keys directly.

Layer	Protection	Subprocessor
Data in transit (dashboard)	HTTPS/TLS (Vercel platform-managed)	Vercel
Data in transit (database)	TLS (Supabase platform-managed)	Supabase
Data at rest (database)	AES-256 (AWS EBS volume encryption)	Supabase / AWS
Data at rest (founder laptop)	FileVault (macOS) / BitLocker (Windows)	Founder laptop OS

Layer	Protection	Subprocessor
Credentials (operator passwords)	Hashed per WorkOS defaults	WorkOS
Credentials (founder secrets)	1Password encryption (end-to-end)	1Password
Credentials (API keys)	Vercel encrypted environment	Vercel

Table 2 — Cryptographic protection scheme by layer.

8. Capacity & Performance Planning

Capacity controls protect the deployment from denial-of-service conditions. Masar Zero relies on subprocessor platform capacity controls:

- **Vercel platform:** Auto-scales serverless functions; per-IP rate-limiting; Always-On DDoS Protection.
- **Supabase:** Connection pooler caps concurrent connections; per-row query cost limits.
- **Station-side router:** Manufacturer-default throughput limits on the external Wi-Fi/cellular uplink.

Masar Zero does not operate its own capacity monitoring. Founders review subprocessor-provided metrics (Vercel analytics, Supabase dashboard) on an ad-hoc basis.

9. ACLs & Data Flow Enforcement

Access control lists enforce explicit data flow policy through subprocessor features: (a) Supabase row-level security policies act as data-flow ACLs at the database layer; (b) Vercel deployment protection acts as origin ACL; (c) WorkOS RBAC acts as application-layer ACL. Masar Zero configures these; the subprocessors enforce them.

10. References

- BAPCO-QR-001 — Master Questionnaire Response §2 (architecture), §7.6, §7.7, §7.9
- MZ-POL-001 — Governance Program
- MZ-POL-003 — Access Control & Identity Management
- Vercel Security — <https://vercel.com/security>
- Supabase Security — <https://supabase.com/security>
- WorkOS Security — <https://workos.com/security>

MZ-POL-006 — Network Security Policy. End of document.